2023 IEEE 24th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) | 978-1-6654-9626-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/SPAWC53906.2023.10304544

# Receiver-Agnostic Radio Frequency Fingerprint Identification via Feature Disentanglement

Hongli Liu[1], Shaoxian Zhu[1], Liu Yang[1], Qiang Li[1,2], Jingran Lin[1,2]
[1]*School of Information and Communication Engineering, University of Electronic Science and Technology of China*
[2]*Laboratory of Electromagnetic Space Cognition and Intelligent Control, Beijing 100089, China*

***Abstract*— The Radio Frequency Fingerprint Identification (RFFI) technique utilizes the subtle and unintentional modulation present in the transmitted RF waveform, caused by the non-ideal characteristics of the device, to uniquely identify an emitter. RFFI has recently gained significant attention, and deep learning (DL) approach has achieved superior performance. In this paper, we study the DL-based RFFI with an emphasis on the fingerprint contamination by receivers. Specifically, during reception, the receivers' fingerprints are inevitably introduced, and different receivers affect emitters' fingerprints differently, thereby degrading recognition performance when the RFFI recognition model is deployed at different receivers. To mitigate the receivers' effect, we propose a receiver-independent recognition model based on the idea of feature disentanglement. With the aid of some carefully designed model structure, loss functions and training strategy, the proposed model learns a feature space, in which the emitters' features are disentangled from the receivers', through multi-receiver's training data. When deploying the model on a new receiver, the emitter-dependent features are exploited to achieve receiver-independent identification. Experiments on two real-world datasets demonstrate that our method significantly outperforms the baseline method without considering fingerprint contamination.**

***Index Terms*—Radio Frequency Fingerprint Identification, receiver-agnostic, domain generalization, feature disentanglement**

## I. Introduction

Radio Frequency Fingerprint Identification (RFFI) is a signal processing technique employed in distinguishing an emitter signal from other signals by comparing extracted distinguishing features called "fingerprint" [1]. The "fingerprint" refers to the unique and subtle modulation characteristics added to the signal by the radio frequency (RF) circuit devices during signal generation process. In the past few years, RFFI has become increasingly prevalent in military and wireless communications for device identification and management.

Traditional RFFI methods primarily depend on the statistical analysis of features extracted from RF signals [2]–[4]. These existing techniques are limited by their reliance on predefined features, which are determined by experts. In recent years, the deep learning (DL) approach has been employed to extract more complex signal characteristics in RFFI, achieving better results than traditional methods [5]–[8].

However, in practice, a model may be trained and deployed with data collected from different receivers. As each receiver also has a distinct "fingerprint", the data from different receivers may have varying distributions. On the other hand, existing RFFI models based on DL require identical distributions for both training and deployment. Consequently, a significant performance loss would be incurred when the RFFI models are trained and deployed with different receivers.

To mitigate the receivers' effect, the recent work [9] uses labeled training data and unlabeled test data to jointly train a model. This approach is more like domain adaptation (DA), i.e., transferring the model knowledge to a new domain with only access to the new domain's data (without label). However, in practice, during the training stage, it is often infeasible to touch the test data. Therefore, a more realistic scenario is that the model is trained with multi-receiver's data and we need to learn a receiver-agnostic RFFI model, which can be directly deployed at any receivers once the training is done. This problem is known as receiver-agnostic RFFI [6], [10], and there is very limited research on this topic in the existing literature.

Receiver-agnostic RFFI shares similarities with domain generalization (DG) [11]. Inspired by DG, we propose a receiver-agnostic RFFI method, known as the Receiver-Independent Emitter Recognition Algorithm based on Information Entropy (RIEI). The key idea of RIEI is to enhance the generalization capabilities of the network by incorporating various losses, including the cross entropy loss, the mutual independence loss and the information entropy loss, to disentangle the emitter-dependent features and the receiver-dependent features. By learning receiver-agnostic features, RIEI effectively mitigates the impact of varying receivers' fingerprints, thereby improving the generalization ability of the model to a new receiver. Compared to the baseline methods that do not consider cross-receiver effects, the proposed method demonstrates a significant accuracy improvement of $9.38\%$ and $20.74\%$ on two real-world datasets.

Our contributions are summarized as follows:

- In this work, we link the receiver-agnostic RFFI to DG, and propose a method inspired by DG and feature disentanglement.
- We propose a new method for receiver-agnostic RFFI that separates emitter-dependent features from receiver-dependent features and allows RFFI on a new receiver without retraining after training on data collected from multiple receivers.

Qiang Li is the corresponding author (Email: lq@uestc.edu.cn). This work is supported by the Natural Science Foundation of China under Grant 62171110.

- Experimenting on two receiver-agnostic RFFI datasets, our proposed method outperforms the baseline method greatly, demonstrating the method's superiority.

## II. Problem Formulation

The multi-source receiver-agnostic RFFI problem can be described as follows. We have a collection of labeled random RF data $\mathcal{S} = \{\mathcal{S}^1, \ldots, \mathcal{S}^M\}$ from $M$ receivers, and $\mathcal{S}^i = \{(x_j^i, y_j^i)\}_{j=1}^{n_i}$ records $n_i$ RF waveform samples $x_j^i$ and its emitter index/label $y_j^i \in \mathcal{Y} = \{1, \ldots, K\}$, where $\mathcal{Y}$ denotes the set of emitters. Due to receiver fingerprint contamination, the joint distribution $P_{XY}$ varies across the receivers, i.e. $P_{XY}^i \neq P_{XY}^j, 1 \leqslant i \neq j \leqslant M$. Our goal is to learn a robust and generalizable classification model $f : \mathcal{X} \to \mathcal{Y}$ that maps the input space to the emitters' index set to achieve a minimum classification error on an unseen test data $\mathcal{S}^{\mathcal{T}} = \{x_j^{\mathcal{T}}\}_{j=1}^{n_{\mathcal{T}}}$ obtained from a new receiver with $P_{XY}^{\mathcal{T}} \neq P_{XY}^i$ for $i \in \{1, \ldots, M\}$. Mathematically, the multi-source receiver-agnostic RFFI problem can be formulated as follows.

$$\min_f \mathbb{E}_{(X,Y)\sim P_{XY}^{\mathcal{T}}}[\mathbb{I}(f(X) \neq Y)], \tag{1}$$

where $\mathbb{E}$ denotes the expectation and $\mathbb{I}(\cdot)$ is the indicator function and $\mathbb{I}(a \neq b) = 1$ if $a \neq b$, and 0 otherwise.

Problem (1) is challenging due to inaccessible of the joint distribution $P_{XY}^{\mathcal{T}}$ or its realization $\mathcal{S}^{\mathcal{T}}$. In the following, we will provide an alternative way to indirectly minimize the classification error probability over $P_{XY}^{\mathcal{T}}$.

## III. Proposed Method

Our proposed method is based on the following intuition. The emitters' fingerprints are superimposed with the receivers' fingerprints at the reception, thereby leading to varied $P_{XY}$ across the receivers. If we are able to train a model from the multi-source data $\mathcal{S}$ that can automatically extract the receiver-independent features, then we may expect that those features should contain only the emitter-dependent characteristics. Consequently, when the data captured by a new receiver is tested, the trained model would highlight the emitter-dependent features and suppress the new receiver's feature. Following the above idea, the proposed RFFI model is shown in Fig. 1. It contains three modules, namely the Feature Extraction and Disentanglement (FED) module, the Emitter Classifier (EC) and the Receiver Classifier (RC), and some judiciously designed loss functions.

- The FED module: It maps the input RF waveform into a high-dimensional feature space such that the emitter-dependent feature and the receiver-dependent feature are disentangled. To this end, some judiciously designed loss functions and training procedure are needed; we will detail this shortly.
- The EC module: It maps the emitter-dependent feature into a $K$-dimensional probability vector, which provides the confidence for predicting which emitter the input signal is from.
- The RC module: It has a similar function as the EC module, except that the input is the receiver-dependent feature and the output is an $M$-dimensional vector, which provides the confidence for predicting which receiver the input signal is from. Note that after training, the RC module is discarded and only the FED and EC modules are used during the test stage.
- The loss functions: Three types of loss functions are devised to train the above modules: 1) The classification entropy (CE) loss, which exploits the emitters' label information and the receivers' label information to minimize classification error probability over $\mathcal{S}$. 2) The mutual independence (MI) loss, which promotes the independence between the emitter-dependent feature and the receiver-dependent feature. 3) The information entropy (IE) loss, which is obtained by calculating the entropy of the EC (RC) module's output when the receiver-dependent (emitter-dependent) feature is input. By maximizing the IE loss, we hope that the EC (R-C) module cannot distinguish the receivers (emitters), and thus the emitters' information is not leaked to the receiver-dependent feature, and vice versa.

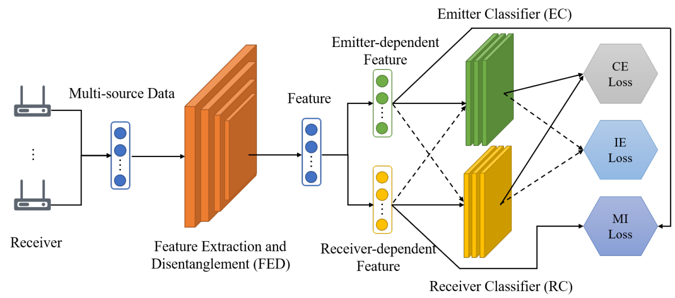


Fig. 1. Network Structure Diagram Based on RIEI.

In the following, we will give more details on how to jointly train the above modules with the aid of the specially designed loss functions.

### A. The Classification Entropy Loss

To proceed, let us denote by $\phi_{\theta_F}(\cdot)$, $\phi_{\theta_E}(\cdot)$ and $\phi_{\theta_R}(\cdot)$ the network function associated with the FED, the EC and RC modules, respectively, and $\theta_F \in \mathbb{R}^{L_F}$, $\theta_E \in \mathbb{R}^{L_E}$ and $\theta_R \in \mathbb{R}^{L_R}$ are the corresponding network parameters. Given an input signal $x$, the FED output is expressed as

$$\phi_{\theta_F}(x) = [\phi_{\theta_F}^{(E)}(x),\ \phi_{\theta_F}^{(R)}(x)],$$

where $\phi_{\theta_F}^{(E)}(x)$ and $\phi_{\theta_F}^{(R)}(x)$ represent the emitter-dependent feature and the receiver-dependent feature, respectively.

By passing the $\phi_{\theta_F}^{(E)}(x)$ to the emitter classifier, we have the output

$$z^E = \delta\left(\phi_{\theta_E}\left(\phi_{\theta_F}^{(E)}(x)\right)\right) \in \mathbb{R}_+^K,$$

where $\delta(\cdot)$ denotes the softmax function, i.e. $[\delta(x)]_i = \frac{e^{x_i}}{\sum_{j=1}^K e^{x_j}}$ for $x \in \mathbb{R}^K$. Notice that $z^E$ lies in a probabilistic simplex satisfying $[z^E]_i \geq 0, \sum_{i=1}^K [z^E]_i = 1$. To encourage the EC correctly classifies the emitters, the cross-entropy loss is resorted:

$$\mathcal{L}_{CE}^{E}=\sum_{m=1}^{M}\sum_{(x,y)\in\mathcal{S}^m}\mathsf{H}(z^E,\mathbb{1}_y), \tag{2}$$

where $\mathsf{H}(\cdot,\cdot)$ denotes the cross-entropy function $\mathsf{H}(p,q)=-\sum_i q_i\log p_i$, and $\mathbb{1}_y\in\mathbb{R}^K$ is a one-hot vector with the $y$-th element being one and zeros otherwise.

Similarly, we can calculate the cross-entropy loss for classifying the receivers via the RC module as follows.

$$\mathcal{L}_{CE}^{R}=\sum_{m=1}^{M}\sum_{(x,y)\in\mathcal{S}^m}\mathsf{H}(z^R,\mathbb{1}_m), \tag{3}$$

where $z^R=\delta\left(\phi_{\theta_R}\left(\phi_{\theta_F}^{(R)}(x)\right)\right)\in\mathbb{R}_+^M$, and $\mathbb{1}_m\in\mathbb{R}^M$.

By combining (2) and (3), we get the classification entropy loss:

$$\mathcal{L}_{CE}=\mathcal{L}_{CE}^{E}+\mathcal{L}_{CE}^{R}. \tag{4}$$

### B. Mutual Independence Loss

To encourage the FED network to learn a receiver-independent feature space, we introduce the following loss to force mutual independence between $\phi_{\theta_F}^{(E)}(x)$ and $\phi_{\theta_F}^{(R)}(x)$.

$$\mathcal{L}_{\mathrm{MI}}=\sum_{m=1}^{M}\sum_{(x,y)\in\mathcal{S}^m}\frac{\left|\langle\phi_{\theta_F}^{(E)}(x),\phi_{\theta_F}^{(R)}(x)\rangle\right|}{\|\phi_{\theta_F}^{(E)}(x)\|\|\phi_{\theta_F}^{(R)}(x)\|}, \tag{5}$$

where $\|\cdot\|$ denotes the Frobenius norm, and $\langle\cdot,\cdot\rangle$ denotes the inner product of two vectors.

The rationale behind the MI loss is as follows. First, from the RF fingerprint generation point of view, the emitters and the receivers have distinct hardwares and information processing flow; therefore, their induced fingerprints in the RF signals should be less correlated. Secondly, with the aid of the high expressiveness of the neural networks, it is expected that in some high-dimensional feature space, the emitters' features would be disentangled from the receivers', owing to the independence of fingerprints at the emitters and the receivers. Therefore, we introduce the MI loss to help the FED network to search for such a feature space.

### C. The Information Entropy Loss

The information entropy loss $\mathcal{L}_{IE}$ is denoted as

$$\mathcal{L}_{IE}=\mathcal{L}_{IE}^{R\rightsquigarrow E}+\mathcal{L}_{IE}^{E\rightsquigarrow R}, \tag{6}$$

where $\mathcal{L}_{IE}^{R\rightsquigarrow E}$ calculates the output entropy when the receiver-dependent feature is input into the emitter classifier, i.e.,

$$\mathcal{L}_{IE}^{R\rightsquigarrow E}=-\sum_{m=1}^{M}\sum_{(x,y)\in\mathcal{S}^m}z^{R\rightsquigarrow E}\odot\log\left(z^{R\rightsquigarrow E}\right), \tag{7}$$

and $\mathcal{L}_{IE}^{E\rightsquigarrow R}$ calculates the output entropy when the emitter-dependent feature is input into the receiver classifier, i.e.,

$$\mathcal{L}_{IE}^{E\rightsquigarrow R}=-\sum_{m=1}^{M}\sum_{(x,y)\in\mathcal{S}^m}z^{E\rightsquigarrow R}\odot\log\left(z^{E\rightsquigarrow R}\right), \tag{8}$$

where $z^{R\rightsquigarrow E}=\delta\left(\phi_{\theta_E}\left(\phi_{\theta_F}^{(R)}(x)\right)\right)\in\mathbb{R}_+^K,\ z^{E\rightsquigarrow R}=\delta\left(\phi_{\theta_R}\left(\phi_{\theta_F}^{(E)}(x)\right)\right)\in\mathbb{R}_+^M$, the $\log(\cdot)$ operation is taken on every element of its argument, and $\odot$ denotes the element-wise product.

We propose to *maximize* the IE loss $\mathcal{L}_{IE}$ to further encourage feature disentanglement. The intuition behind IE loss is as follows. Take $\mathcal{L}_{IE}^{R\rightsquigarrow E}$ for example. The larger $\mathcal{L}_{IE}^{R\rightsquigarrow E}$ implies the larger uncertainty of the prediction made by the emitter classifier. Suppose that the emitter classifier has been well trained to differentiate emitters accurately, the larger uncertainty of the prediction means that the prediction vector $z^{R\rightsquigarrow E}$ tends to more uniformly distributed; therefore the emitter classifier cannot accurately predict the emitters from the receiver-dependent feature. From information theoretic point of view, the receiver-dependent feature should contain little information about the emitters.

### D. Training

The overall loss function is given by

$$\mathcal{L}(\theta_F,\theta_E,\theta_R)=\mathcal{L}_{CE}+\lambda_1\mathcal{L}_{MI}-\lambda_2\mathcal{L}_{IE}, \tag{9}$$

where $\lambda_i\in\mathbb{R}_+,\ i=1,2$ are hyperparameters. A straightforward way to train $(\theta_F,\theta_E,\theta_R)$ is using stochastic gradient descent (SGD) to update the parameters simultaneously. However, we numerically found that SGD is not stable and usually results in poor performance for our considered loss. We guess the reason is that the feature disentanglement is much harder than conventional feature extraction task, and the former is more sensitive to the update of the classifier. Therefore, the update of the classifier and the FED should be carefully balanced. To this end, we propose an alternating training with intermediate updating scheme to optimize the network parameters. Specifically, it includes the following steps. Randomly initialize $(\theta_F^0,\theta_E^0,\theta_R^0)$ and alternately update the classifier and the feature extractor:

1) Update the classifier:

$$\theta_E^{\ell+1}\leftarrow\theta_E^{\ell}-\eta_E^{\ell}\frac{\partial\mathcal{L}_{CE}}{\partial\theta_E}, \tag{10a}$$

$$\theta_R^{\ell+1}\leftarrow\theta_R^{\ell}-\eta_R^{\ell}\frac{\partial\mathcal{L}_{CE}}{\partial\theta_R}, \tag{10b}$$

$$\theta_F^{\ell+\frac{1}{2}}\leftarrow\theta_F^{\ell}-\eta_F^{\ell}\frac{\partial\mathcal{L}_{CE}}{\partial\theta_F}, \tag{10c}$$

2) Update the feature extractor for fixed $(\theta_E^{\ell+1},\theta_R^{\ell+1})$:

$$\theta_F^{\ell+1}\leftarrow\theta_F^{\ell+\frac{1}{2}}-\eta_F^{\ell}(\lambda_1\frac{\partial\mathcal{L}_{MI}}{\partial\theta_F}-\lambda_2\frac{\partial\mathcal{L}_{IE}}{\partial\theta_F}), \tag{11}$$

where $\ell$ is the iteration index and $\eta^{\ell}$ is the step-size for the $\ell$-th iteration.

We should mention that when updating the classifier, we have also performed an intermediate update for the feature extractor in (10c). This intermediate update is crucial to stabilize the learning status, since $\theta_F^{\ell+\frac{1}{2}}$ ties the update of classifier and the feature extractor.

TABLE I
THE OVERALL STRUCTURE OF THE WISIG AND HACKRF DATASETS.

| Parameter | Wisig | HackRF |
|---|---|---|
| Emitter | 6 | 4 |
| Receiver | 6 | 3 |
| Input sample dimension | $2 \times 256$ | $2 \times 224 \times 224$ |
| Training samples per receiver | 14400 | 3200 |
| Testing samples per receiver | 4800 | 800 |

TABLE II
COMPARISON OF RECOGNITION ACCURACY (%) ON THE HACKRF DATASET.

| Model | HackRF 1 | HackRF 2 | HackRF 3 |
|---|---|---|---|
| RIEI | $53.41 \pm 5.21$ | $79.21 \pm 6.79$ | $73.22 \pm 2.60$ |
| Baseline | $34.56 \pm 2.44$ | $54.81 \pm 2.44$ | $54.24 \pm 6.76$ |

# IV. EXPERIMENT

This section presents the results of RIEI on two datasets, namely the Wisig dataset [12] and the HackRF dataset[1]. We demonstrate the benefits of RIEI by comparing it with the baseline approach, which solely relies on cross-entropy loss for training of the FED and RC. The details of the datasets and models are as follows.

*1) Dataset:* The Wisig dataset is constructed using WiFi nodes as emitters, emitting signals to another WiFi access point, and using multiple USRPs to receive WiFi signals. Both emitters and receivers are placed indoors in a rectangular configuration with a total of 174 WiFi signal emitters and 41 USRP receivers. The location of receivers is identified by their horizontal and vertical coordinates. For instance, the receivers situated at position $(1, 1)$ in both horizontal and vertical planes are denoted as receiver $Rx_{(1-1)}$. According to [12], the original Wisig signal is preprocessed by eliminating portions without signal and followed by channel equalization. After data preprocessing, $80\%$ of the valid data is used for training and the remaining $20\%$ for testing. The dataset structure is presented in Table I.

The HackRF dataset is generated and collected using the open-source and cost-effective HackRF platform with carrier frequency at 2.5GHz, bandwidth 2MHz and BFSK modulation. Dataset is randomly partitioned for training and testing, similar to the Wisig dataset. Table I provides an overview of the HackRF dataset structure.

*2) Model setup:* The FED employs Residual Networks (ResNets) [13] for both the Wisig and HackRF datasets. Specifically, ResNet1D-18 and ResNet50 with attention are utilized as FED modules for the Wisig and HackRF datasets, respectively. Both the EC and RC are a 3-layer fully connected network. Furthermore, the baseline method uses the same FED and EC modules as the proposed method.

## A. Comparison with Baseline

In the Wisig and HackRF experiments, we feed the models with signals and time-frequency diagrams obtained through Short-time Fourier Transform (STFT), respectively. The experimental design entails training the model on data from two receivers, followed by testing on data from a third receiver. To assess the model, we take into account the mean and standard deviation of the classification accuracy for the last 10 epochs.

In Table II, we present the classification accuracy achieved by RIEI and baseline approaches on the HackRF dataset. Data from the HackRF $i$ column pertains to testing using data from the HackRF $i$ receiver and training using data from the remaining two receivers. The results clearly depict a penchant for RIEI to outshine the baseline in terms of recognition accuracy, with an average improvement of 20.74%.

Table III displays the test results on the Wisig dataset. RIEI outperforms the baseline approach, resulting in an average recognition accuracy improvement of 9.38%.

## B. Ablation Studies

Ablation studies are conducted to investigate the impact of information entropy loss and mutual independence loss on the Wisig and HackRF datasets. As presented in Table IV, the notation (h$i$, h$j$)$\rightarrow$h$k$ indicates that the model is trained using data from receivers HackRF $i$ and HackRF $j$, and tested using data from receiver HackRF $k$, while $(Rx_{(x-y)}, Rx_{(x'-y')}) \rightarrow Rx_{(x''-y'')}$ signifies an identical process. Results indicate that using either information entropy loss or mutual independence loss alone results in an improved recognition accuracy of 14.79% and 14.09%, respectively, compared to models without both loss functions. Combining both loss functions results in a significant increase in recognition accuracy by 18.16% compared to the baseline model. Furthermore, using both loss functions has better results compared to using only one of them, with accuracy improved by 3.38% and 4.08%, respectively. It is therefore concluded that both information entropy loss and mutual independence loss are effective for optimizing network performance.

## C. t-SNE Visualization

In this subsection, we demonstrate that the proposed RIEI model indeed disentangles the features of emitters and receivers by drawing the scatter plot of the dimension-reduced emitter-dependent features. For illustration, we use HackRF 1 and HackRF 2 to train the model and test it on HackRF 3. Fig. 2(a)-Fig. 2(c) show the emitter-dependent features obtained by t-distributed Stochastic Neighbor Embedding (t-SNE) with different colors representing different emitters. As seen, the emitter-dependent features of HackRF 3 follow a similar distribution as HackRF 1 and HackRF 2, which implies that the proposed model can effectively mitigate the effect of receivers and make the feature distributed consistently over different receivers. By contrast, Fig. 2(d) presents the t-SNE visualization obtained by the baseline method. Clearly, different emitters' features are densely overlapped due to the ignorance of the receiver effect.

# V. CONCLUSION

This paper has addressed the fingerprint contamination problem, which arises from different characteristics of re-

[1]The Wisig data is a public available dataset, which can be downloaded at https://cores.ee.ucla.edu/downloads/datasets /wisig/. The HackRF dataset is produced by using the software defined radio HackRF One as the emitters and receivers.

TABLE III

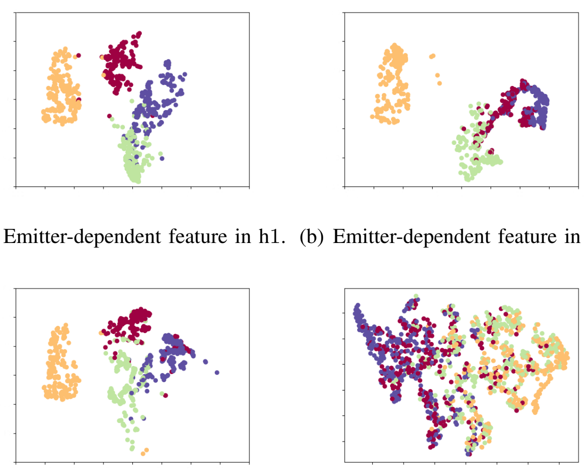
COMPARISON OF RECOGNITION ACCURACY (%) FOR TWO RECEIVERS TRAINING ANOTHER RECEIVER TESTING IN THE WISIG DATASET.

| Training data receivers | Test data receiver | Baseline accuracy | RIEI accuracy | Improved accuracy |
|---|---|---|---|---|
| $Rx_{(1-1)}$, $Rx_{(7-7)}$ | $Rx_{(1-19)}$ | $70.39 \pm 1.37$ | $77.88 \pm 2.23$ | 7.49 |
| $Rx_{(1-1)}$, $Rx_{(8-8)}$ | $Rx_{(1-19)}$ | $62.85 \pm 0.75$ | $79.43 \pm 1.66$ | 16.58 |
| $Rx_{(1-1)}$, $Rx_{(14-7)}$ | $Rx_{(1-19)}$ | $64.78 \pm 0.72$ | $66.09 \pm 0.67$ | 1.31 |
| $Rx_{(7-7)}$, $Rx_{(8-8)}$ | $Rx_{(1-19)}$ | $69.16 \pm 1.26$ | $70.51 \pm 3.53$ | 1.35 |
| $Rx_{(7-7)}$, $Rx_{(14-7)}$ | $Rx_{(1-19)}$ | $65.34 \pm 0.54$ | $77.35 \pm 1.53$ | 12.01 |
| $Rx_{(8-8)}$, $Rx_{(14-7)}$ | $Rx_{(1-19)}$ | $62.51 \pm 0.80$ | $75.48 \pm 1.21$ | 12.97 |
| $Rx_{(1-1)}$, $Rx_{(1-19)}$ | $Rx_{(14-7)}$ | $66.54 \pm 1.31$ | $71.91 \pm 2.08$ | 5.37 |
| $Rx_{(1-1)}$, $Rx_{(7-7)}$ | $Rx_{(14-7)}$ | $60.49 \pm 1.57$ | $68.33 \pm 2.37$ | 7.84 |
| $Rx_{(1-1)}$, $Rx_{(8-8)}$ | $Rx_{(14-7)}$ | $61.10 \pm 0.56$ | $73.54 \pm 1.27$ | 12.45 |
| $Rx_{(1-19)}$, $Rx_{(7-7)}$ | $Rx_{(14-7)}$ | $62.97 \pm 1.89$ | $73.52 \pm 3.15$ | 10.56 |
| $Rx_{(1-19)}$, $Rx_{(8-8)}$ | $Rx_{(14-7)}$ | $65.61 \pm 2.41$ | $72.05 \pm 2.71$ | 6.43 |
| $Rx_{(7-7)}$, $Rx_{(8-8)}$ | $Rx_{(14-7)}$ | $55.31 \pm 1.48$ | $73.46 \pm 2.00$ | 18.14 |

TABLE IV

COMPARISON OF RECOGNITION ACCURACY (%) OF ABLATION STUDIES.

| Components | | (h1,h2) | (h1,h3) | (h2,h3) | ($Rx_{(1-1)}$, $Rx_{(8-8)}$) | ($Rx_{(7-7)}$, $Rx_{(14-7)}$) | ($Rx_{(7-7)}$, $Rx_{(8-8)}$) |
|---|---|---|---|---|---|---|---|
| IE | MI | $\to$ h3 | $\to$ h2 | $\to$ h1 | $\to Rx_{(1-19)}$ | $\to Rx_{(1-19)}$ | $\to Rx_{(14-7)}$ |
| $\times$ | $\times$ | $54.24 \pm 6.76$ | $54.81 \pm 2.44$ | $34.56 \pm 2.44$ | $62.85 \pm 0.75$ | $65.34 \pm 0.54$ | $55.31 \pm 1.48$ |
| ✓ | $\times$ | $67.30 \pm 14.17$ | $72.14 \pm 10.14$ | $52.77 \pm 6.98$ | $76.94 \pm 0.86$ | $74.53 \pm 0.76$ | $72.15 \pm 1.23$ |
| $\times$ | ✓ | $72.40 \pm 4.10$ | $72.90 \pm 4.90$ | $52.25 \pm 4.63$ | $77.74 \pm 1.41$ | $68.12 \pm 2.32$ | $68.21 \pm 1.56$ |
| ✓ | ✓ | $\mathbf{73.22 \pm 2.60}$ | $\mathbf{79.21 \pm 6.79}$ | $\mathbf{53.41 \pm 5.21}$ | $\mathbf{79.43 \pm 1.66}$ | $\mathbf{77.35 \pm 1.53}$ | $\mathbf{73.46 \pm 2.00}$ |



(a) Emitter-dependent feature in h1. (b) Emitter-dependent feature in h2.

(c) Emitter-dependent feature in h3. (d) Baseline.

Fig. 2. Visualization with t-SNE of emitter-dependent features. Different colors represent the features of different emission radiation sources.

ceivers, in radio frequency fingerprint identification (RFFI). A receiver-agnostic RFFI model is proposed by disentangling the receivers' features from the emitters'. To this end, we have developed three specially designed losses, namely the cross-entropy loss, the mutual independence loss and the information entropy loss, to encourage the model to learn a receiver-agnostic feature space, thereby mitigating the receivers' effects. Experimental results demonstrate that our proposed method outperforms the traditional baseline method with accuracy improvement of 9.38% and 20.74% on the Wisig dataset and HackRF dataset, respectively.

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